INTEGUMENTARY SYSTEM

COMPREHENSIVE STUDY NOTES

1. Structure and Functions of Skin

- Integumentary system contains skin, hair, glands, hands, nails.
- Integumentary system which cover the outside of the body...
- The Average thickness of Skiness is 1 to 2 mm. Skin is the largest organ in the body.
- Skin is physical barrier between External and Internal Environment.
- Skin gives protection.
- At soles of foot, palm of hand, Inter scapular region thickness is "5mm resp". *"If the skin is about 5mm thick, that area has no hair and no glands."*
- Eye Lids and penis skin, thickness is "0.1 mm".

2. Layers of Skin

Skin is divided into 2 types:

- **Epidermis**
- **Dermis**

3. Epidermis

- This is made up of Stratified Epithelium.
- This is the outer most layer.
- It does not have blood vessels.
- It Supply natrition by dermal capillary... to Epidermis.

Layers of Epidermis (From Deep to Superficial):

- **Stratum Germinativum (Basal Layer)**:
  - This is the thickest layer.
  - In stratum germinative, New cells by Mitotic division.
  - Here New cells called as "Keratinocytes".
  - Chromosomes are normally not visible only. They become visible during cell division called Mitotic division.
  - In between the Keratinocytes there are specialised melanocytes which core some scattered called. Function of melanocytes to produce "melanin" mutlsdlige buifitarts to colour pigment.
- **Stratum Spinosum (Prickle Cell Layer)**:
  - It also called as "prickle cell layer".
  - It contains "Rhomboid cells".
  - It has two spine Like Projection they are 1) superficial, 2) deeper.
- **Stratum Granulosum**:
  - It contains 2-5 rows of flattened epitheal cells.
  - These cells are dead having granules with Protein "Keratohyalin".

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    - This keratohyalin act as preassor for "Keratin".

- **Stratum Lucidum**:
    - It contains mainly flattened dead cells having shiny cytoplasm.
    - Lucid means "clear".

- **Stratum Corneum**:
    - Known as Horny Layer.
    - cells called as "Corneocytes".
    - The cytoplasm Here is flattened with protein Keratin.

4. Dermis

- It is the most Inner most layer connective tissue layer.
- It connects outer Epidermis and subqutoncous tissue.
- It contains dense stout collagen fibres having Elasticity nature.
- collagen fibres can able to storage & water holding capacity.
- dermis contains Enzymes called as "collagenase". The Enzyme collagenase responsible for "wound healing".

Subdivisions of Dermis:

- **Superficial Papillary Layers**:
    - This layer projects into the epidermis.
    - It contains blood vessels and nerve fibres.
    - also contains finger like projections Aris from this layer called "dermal papillae".

- **Deep Reticular Layer**:
    - It contains Elastic and reticular fibers.
    - Immediately below the dermis called as "Subcutaneous tissue".
    - This tissues connect the skin with Internal organs of the body.

Appandages of skin:

- These are the projecting parts with distinct apperance & function.

5. Colour of the Skin

- **Pigmentation of skin**:
    - Skin contain a brown coloured pigment which is called a "melanin".
    - Melanin Synthesized by "melanocytes".
    - Melanin is a protein. It is Synthesized from an Aminoacid with Tyrosine.
    - defeciency of melanin content in a person called as Albinism.

- **Haemoglobin content**:
    - The Amount and the nature of blood that Circulates in the cataneous blood vessels plays Important role in determaining the skin colour.
    - *Eg:* If we take pale color: Low heamoglobin Content.
    - If the colour is pink: cutaneous vasodilatation.

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    - If the colour is blue: Cyanosis.

6. Glands

Skin contains two types of glands:

1. Sebaceous glands
2. Sweat glands

1. Sebaceous Glands:

- These glands present at the side of hair follicle.
- are oval or round shaped.
- These sebaceous gland develop from hair follicle on thick skin.
- Sebaceous glands are absent on thick skin.
- The Alveoli of the gland are lined by stratified Epithelial cells.
- These glands Secreates oily substance Known as "sebum".
- These glands opens through hair follicle by a duct. Sometimes directly opens to the Epithelial Surface.
- Four places directly opens to the epithelial surface they are: face, Lips, nipple area, Labia minora.
- **Composition & Functions of Sebum**:
    - The sebum contains free fatty acids, waxes, Triglycerides, paraffin lipids.
    - sebum is having Antibacterial / Antifungal activity.
    - it prevents the Entry of microrganisms into the body.
    - Because of Lipids in ssebum it keeps the skin smooth/oil.
    - the Lipids prevents the heat loss from the body.
- **Activation & Conditions**:
    - They are Inactive until puberty.
    - When the skin become dry called Exema.
    - Due to Increased seceretory of sex hormones dehydroepiandrosterone then ssebaseous glands are becoming active suddenly.
    - which leads to development of Acne on face.
    - Acne means Localised inflammatry condition of skin which is characterized by pimples in face, chest, neck.
    - The common form of Acne occuring during Adolscence called Acne vulgaries.

2. Sweat Glands:

These gland mainly two types: 1. Eccrine, 2. Apocrine.

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Feature	Eccrine Glands	Apocrine Glands
Distribution	Distribute throughout the body	Only certain area like Axilla, pubis, Areola, umbilicus
Duct Opening	Open through a sweat pore	Open through hair follicle
Lifespan Activity	Functioning throughout the life	Start functioning after puberty
Temp. Regulation	Involve in temperature regulation	Didn't Involve in temperature regulation
Secretion Triggers	Release secretion when Increased temperature and also Increased emotional conditions	Release secretion during Emotional conditions
Nervous Control	Under nervous control (Acetyl choline controls the Sweat glands)	Under hormonal control (Adrenaline controls Sweat glands)
Secretion Type	Clear wattery sweat, contains water, Lactic acid, urea, NaCl	Milky / sticky / thicky, odowiless when fresh (smell starts after microbial decomposition)

7. Functions of the Skin

- **Protection**:
  - The first function of skin is protection.
  - Skin Secreates an Enzyme called Lysozyme.
  - Lysozyma destroys the bacteria.
  - Keratinised Stratified squamous epithelium of Epidermis is responsible for the protection.
  - The keratinocytes secretes antimicrobial substances called β-defensin which prevents the entry of microbes.
  - The Keratinocytes also secretes some Substance like Inflammatory mediators, INF-α, cytokines. These also helps in wound healing.
  - Skin also protection from mechanical blow.
  - Skin also protects from UV rays with help of melanin.
- **Sensory Function**:
  - skin contains small sensitive nerve Endings Converted in to cutaneous receptors.
  - They are Stimulated by sensation of touch, pain, pressure, temperature.
- **Storage Function**:
  - skin stores fat, water, chlorides.
- **Synthetic Function**:
  - Vitamin D_3 is Synthesized by skin by the action of UV rays of Sunlight on cholestrol.
- **Regulation of body temperature**:
  - Skin plays an Important role in the regulation of body temperature.
  - Sweat from Sweat glands plays an active role in heat loss by Secreating Sweat.

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- The sebum from Sebaceous gland prevents the heat loss during cold Environment.

- **Regulation of water & Electrolyte balance**:
  - Skin regulates water and Electrolyte balance by Excretion Salt or water in the form of sweat.

- **Excreatory function**:
  - The Skin Excretes waste materials like water, urea, Salts, etc.

- **Secreatory function**:
  - Skin Secrets sweat from the Sweat glands and sebum from Sebaceous glands.
  - By secreating sweat it regulates the body temperature, Electrolytic & water balance.
  - By secreating sebum it keeps the skin smooth.

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